

1. Introduction

Overview

Anti-Wallhack Visibility System is a server-authoritative visibility validation solution designed for Unity multiplayer games.

Unlike traditional client-side visibility systems, visibility decisions are performed independently of the rendering state, preventing players from gaining information through wallhack-style cheats or unauthorized visibility modifications.

The system validates whether a target should be considered visible by combining distance checks, field-of-view validation, visibility sampling, and physics-based line-of-sight tests.

This approach allows developers to synchronize only the information that players are legitimately allowed to see while maintaining a fair multiplayer environment.

The system is suitable for:

- Competitive PvP games
 - Tactical shooters
 - Extraction shooters
 - Survival multiplayer games
 - Stealth-based games
 - AI perception systems
 - Server-authoritative multiplayer architectures
-

Key Features

Server-Authoritative Visibility Validation

Visibility decisions are calculated independently of the client, reducing the possibility of wallhack exploits and unauthorized information access.

Observer-Based Detection

Each observer evaluates visibility from its own perspective using configurable field-of-view and distance settings.

Visibility Sampling System

The system combines fixed and dynamic sampling lines to determine whether a target is genuinely visible.

Broad-Phase Optimization

Early rejection checks reduce the number of expensive visibility calculations, improving scalability in multiplayer environments.

Velocity Compensation

Target visibility can be expanded based on movement speed, helping reduce inconsistencies during fast movement and corner peeking situations.

Multiplayer Ready

The architecture is designed to integrate with popular Unity networking frameworks and server-authoritative multiplayer solutions.

Debug Visualization

Built-in gizmos allow developers to visualize visibility calculations directly inside the Unity Editor.

Free Edition Features

The Free Edition includes:

- ObserverManager
- PlayerObserver
- PlayerVisibilityDetector
- PlayerBox
- Fixed Visibility Sampling
- Dynamic Visibility Sampling
- Velocity Compensation
- 180° Observer Mode
- 360° Observer Mode
- Observer Update Rate Control
- Networking Adapter Components
- Debug Visualization Tools

Some advanced systems are available exclusively in the Professional Edition.

For the most up-to-date information, examples, and framework integrations, refer to the online documentation.

Online Documentation:

<https://gustavobianco277.github.io/Anti-Wallhack-Visibility-System/>

2. Quick Start

Installation

Import the Anti-Wallhack Visibility System package into your Unity project.

After importing the package, create a test scene and configure the required components.

Step 1 – Create an ObserverManager

Create an empty GameObject in your scene and add the:

ObserverManager

component.

The ObserverManager is responsible for:

- Registering observers
- Registering players
- Managing visibility states
- Dispatching visibility events
- Executing visibility updates

Only one ObserverManager should exist in a scene.

Step 2 – Configure a Player Observer

Create or select the object that represents the player's view.

In most projects, this is the camera or a transform that follows the camera orientation.

Add:

PlayerObserver

Configure:

- Observer Angle
- Camera FOV
- View Distance
- Aspect Ratio
- Detection Mode

The observer determines which targets are evaluated for visibility.

Step 3 – Configure a Target Player

Select the player character that should be detected by other observers.

Add:

PlayerVisibilityDetector

and configure a:

PlayerBox

component.

The PlayerBox defines the visibility volume used by the detection system.

Step 4 – Add Network Adapter Components

For multiplayer projects, use the provided adapter components:

NetworkPlayerObserver
NetworkPlayerVisibilityDetect
or

These components initialize the visibility system and forward update calls to the core visibility components.

By default, the adapters use Unity's Update loop and are intended as examples that can be adapted to your networking framework.

Step 5 – Run the Scene

When the scene starts:

1. ObserverManager initializes.
 2. Observers register automatically.
 3. Players register automatically.
 4. Visibility validation begins.
 5. Visibility events are generated as players become visible or hidden.
-

Recommended First Test

Create two cubes separated by a wall.

Assign:

PlayerObserver

to one object and:

PlayerVisibilityDetector
PlayerBox

to the other.

Move the observer around the wall and observe visibility changes through the debug gizmos.

This is the fastest way to verify that the system is functioning correctly before integrating it into a multiplayer project.

3. ObserverManager

Overview

ObserverManager is the central coordinator of the visibility system.

Its primary responsibility is to manage observers, players, visibility states, and visibility update execution.

All visibility calculations pass through ObserverManager, making it the core component responsible for controlling how often visibility validation occurs and how visibility information is distributed throughout the system.

Only one ObserverManager should exist per scene.

Responsibilities

ObserverManager is responsible for:

- Registering observers
- Registering players
- Managing visibility states
- Dispatching visibility events
- Controlling update frequency
- Coordinating visibility validation

The component automatically tracks all registered observers and visibility targets during runtime.

Automatic Registration

When initialized, the following components automatically register themselves with ObserverManager:

PlayerObserver
PlayerVisibilityDetector

Manual registration is not required under normal circumstances.

Visibility State Management

ObserverManager stores visibility states for every observer–target pair.

Whenever a visibility change occurs, the manager automatically dispatches a visibility event.

Examples:

Visible → **Hidden**
Hidden → **Visible**

Visibility events are only triggered when the state actually changes, preventing unnecessary event spam.

Update Rate Control

One of the most important responsibilities of ObserverManager is controlling how frequently visibility updates are executed.

This allows developers to balance accuracy and performance depending on the requirements of their game.

Unlimited Mode

Call Rate Mode: Unlimited

Visibility updates execute as frequently as possible.

Advantages:

- Maximum responsiveness
- Immediate visibility updates
- Best accuracy

Disadvantages:

- Higher CPU usage
- Less scalable for large player counts

Recommended for:

- Small multiplayer games
 - Local testing
 - Debugging
-

Limited Mode

Call Rate Mode: Limited

Visibility updates execute at a fixed frequency.

Advantages:

- Reduced CPU usage
- Better scalability
- Predictable update cost

Disadvantages:

- Slightly increased visibility latency

Recommended for:

- Dedicated servers
- Medium to large multiplayer games
- Production environments

Recommended Update Rates

Typical values:

Update Rate	Usage
16 TPS	Large-scale games
32 TPS	Recommended default
64 TPS	Competitive environments

The included benchmark tests were executed using:

32 visibility updates per second

which provides a good balance between responsiveness and performance.

Performance Considerations

ObserverManager directly affects the scalability of the visibility system.

The following factors influence overall performance:

- Number of observers
- Number of players
- Update rate
- View distance
- Detection mode
- Visibility sampling density

For best results:

- Use Limited mode in production environments
 - Avoid unnecessarily high update rates
 - Tune visibility settings according to gameplay requirements
-

Debugging

ObserverManager provides runtime information that can be used to verify:

- Registered observers
- Registered players
- Visibility state changes
- Update execution

This information is useful when diagnosing visibility issues during development.

Best Practices

- Use a single ObserverManager per scene
- Prefer Limited mode for production builds
- Start with 32 updates per second
- Profile performance before increasing update frequency
- Keep visibility settings proportional to the scale of the game

ObserverManager acts as the foundation of the visibility system and should always be configured before setting up observers and visibility targets.

4. PlayerObserver

Overview

PlayerObserver represents the perspective from which visibility is evaluated.

In most multiplayer games, the observer corresponds to the player's camera, but it can also be used for AI agents, security cameras, spectators, or any entity capable of observing the game world.

The observer is responsible for determining which targets should be evaluated before visibility sampling begins.

This stage acts as the first major filtering layer of the visibility pipeline and helps reduce unnecessary calculations.

Responsibilities

PlayerObserver is responsible for:

- Distance validation
- Field-of-view validation
- Broad-phase rejection
- Visibility target selection
- Observer prediction
- Visibility event handling

Only targets that pass the observer validation stage proceed to visibility sampling.

Observer Angle Modes

180° Mode

Max Observer Angle: 180°

This mode simulates a traditional camera-based field of view.

Only targets located inside the observer's viewing direction are considered for visibility validation.

Advantages:

- Better performance
- More realistic player perception
- Recommended for most games

Recommended for:

- FPS games
 - TPS games
 - Tactical shooters
 - Survival games
-

360° Mode

Max Observer Angle: 360°

This mode removes directional restrictions and allows visibility checks in every direction around the observer.

Advantages:

- Useful for AI systems
- Useful for top-down games
- Useful for omnidirectional detection

Disadvantages:

- More visibility candidates
- Higher processing cost

Recommended for:

- AI perception systems
 - Strategy games
 - Top-down multiplayer games
-

Camera Settings

Camera FOV Angle

cameraFOVAngle

Defines the base field of view used during visibility calculations.

Larger values:

- Increase visible area
- Increase visibility candidates

Smaller values:

- Reduce visible area
- Improve performance

Typical values:

**60° - Competitive
shooters 75° - General
gameplay 90° - Wide
field of view**

View Distance

viewDistance

Defines the maximum distance at which targets can be considered.

Targets outside this range are immediately rejected.

Reducing view distance can significantly improve performance in large multiplayer environments.

Aspect Format

**4:3
16:9
16:10
21:9
32:9**

The selected aspect ratio is used to calculate the horizontal visibility limits of the observer.

Using the correct aspect ratio ensures that visibility calculations match the player's actual camera configuration.

Broad-Phase Optimization

Broad Phase FOV Multiplier

broadPhaseFOVMultiplier

Before performing precise visibility validation, the system executes a broad-phase test.

This expanded FOV allows the observer to quickly reject targets that are clearly outside the viewing area.

Benefits:

- Reduces expensive calculations
- Improves scalability
- Reduces visibility sampling workload

Typical values:

1.4 - 1.8

Recommended default:

1.7

Detection Modes

PlayerObserver supports multiple methods for selecting visibility candidates.

Closest Point

DetectionMode: ClosestPoint

The system evaluates visibility using the closest point on the target's visibility volume.

Advantages:

- Fastest method
- Lowest CPU cost

Recommended for:

- Large player counts
- Performance-sensitive environments

Bounds Corners

DetectionMode: BoundsCorners

The system evaluates multiple points around the target bounds.

Advantages:

- Higher precision
- Better edge visibility handling

Disadvantages:

- More calculations

Recommended for:

- Competitive games
 - High precision environments
-

Observer Prediction

The Free Edition includes movement-based observer prediction.

This system helps compensate for rapid player movement and reduces inconsistencies during corner peeking situations.

Velocity Prediction

The observer can temporarily offset its virtual position based on movement velocity.

Benefits:

- Smoother visibility transitions
 - Better handling of fast movement
 - Reduced corner-peeking inconsistencies
-

Prediction Axis Modes

Supported modes:

Lateral
Frontal
LateralAndFrontal

These modes determine which movement directions contribute to prediction.

Prediction Weights

The following settings control prediction influence:

lateralWeight
frontalWeight
velocityPredictionScale
maxPredictionDistance

Increasing these values improves movement compensation but may slightly increase visibility permissiveness.

Visibility Events

PlayerObserver automatically receives visibility updates from ObserverManager.

Whenever visibility changes:

Visible → Hidden
Hidden → Visible

the observer updates its current visibility state.

This information can be used for:

- Rendering control
 - Network synchronization
 - Gameplay systems
 - Debugging
-

Performance Recommendations

For most multiplayer games:

180° Observer Mode
BoundsCorners
32 Updates Per Second
16:9 Aspect Ratio

provides an excellent balance between accuracy and performance.

Large-scale games should consider:

180° Observer Mode
ClosestPoint
Reduced View Distance
Limited Update Rate

to maximize scalability.

Best Practices

- Match the observer to the player's camera orientation
- Use 180° mode whenever possible
- Tune view distance according to map size
- Use ClosestPoint when prioritizing performance
- Use BoundsCorners when prioritizing accuracy
- Profile performance before increasing visibility ranges

PlayerObserver serves as the first filtering stage of the visibility system and plays a critical role in balancing visibility accuracy and server performance.

5. PlayerVisibilityDetector

Overview

PlayerVisibilityDetector represents a visibility target within the Anti-Wallhack Visibility System.

While PlayerObserver determines which targets should be evaluated, PlayerVisibilityDetector is responsible for performing the actual visibility validation through physics-based sampling and line-of-sight checks.

This component is attached to every player or entity that can be detected by observers.

The detector works together with PlayerBox to generate visibility samples and determine whether a target should be considered visible.

Responsibilities

PlayerVisibilityDetector is responsible for:

- Visibility sampling
- Line-of-sight validation
- Fixed line generation
- Dynamic line generation
- Visibility stabilization
- Velocity compensation
- Visibility state tracking
- Observer management

This component performs the most detailed stage of the visibility pipeline.

PlayerBox Integration

PlayerVisibilityDetector requires a PlayerBox component.

PlayerBox defines the visibility volume used during visibility sampling.

The detector uses the PlayerBox dimensions to:

- Generate visibility samples
- Calculate visibility points
- Perform line-of-sight validation
- Apply velocity compensation

For best results, the PlayerBox should closely match the visible shape of the character.

Visibility Sampling System

The detector uses two sampling methods:

Fixed Lines
Dynamic Lines

These systems work together to improve visibility accuracy while maintaining performance.

Fixed Lines

Fixed lines are static visibility samples distributed across the visibility volume.

They provide consistent visibility validation and ensure that important areas of the target are always checked.

Advantages:

- Stable results
- Consistent visibility validation
- Low computational cost

Disadvantages:

- May miss small visibility openings
 - Less adaptive than dynamic sampling
-

Fixed Line Configuration

verticalFixedLineCount
horizontalFixedLineCount

Increasing these values:

- Improves accuracy
- Increases CPU usage

Reducing these values:

- Improves performance
 - Reduces visibility precision
-

Dynamic Lines

Dynamic lines move across the visibility volume over time.

This allows the system to gradually sample areas that fixed lines may not cover.

Advantages:

- Improved visibility coverage
- Better corner detection
- Improved detection through narrow openings

Disadvantages:

- Slightly higher computational cost

Dynamic Line Configuration

verticalDynamicLineCount
horizontalDynamicLineCount
dynamicLineSpacing
dynamicLineSpeed

Dynamic Line Speed

dynamicLineSpeed

Controls how quickly dynamic sampling moves across the visibility volume.

Higher values:

- Faster coverage
- Improved responsiveness

Lower values:

- Reduced CPU usage
- Slower visibility updates

Recommended starting values:

5 - 15

depending on game requirements.

Grid Subdivision

The visibility volume is subdivided into rows and columns.

This subdivision determines how visibility samples are distributed.

Configuration:

horizontalColumnCount
verticalRowCount

Higher values:

- More visibility samples
- Better accuracy

Lower values:

- Better performance
 - Fewer visibility checks
-

Visibility Stabilization

State Stabilize Time

stateStabilizeTime

Visibility changes near walls or corners can sometimes fluctuate rapidly.

State stabilization introduces a small delay before visibility changes are confirmed.

Benefits:

- Reduces flickering
- Improves visual consistency
- Creates smoother visibility transitions

Typical values:

0.05 - 0.25 seconds

Velocity Compensation

The Free Edition includes velocity-based visibility compensation.

Fast-moving players may partially expose themselves before their visibility volume fully enters the observer's field of view.

Velocity compensation expands the visibility volume based on movement speed.

Benefits:

- Fairer visibility validation
- Improved corner peeking behavior
- Better synchronization between movement and visibility

Expanded Visibility Volume

As movement speed increases:

Movement Speed ↑
↓
Visibility Volume Expansion ↑

This allows fast-moving targets to be detected more consistently.

The expansion amount can be configured through PlayerBox settings and movement compensation parameters.

Observer Tracking

PlayerVisibilityDetector maintains a list of active observers.

Observers are automatically added and removed by the visibility system.

This allows the detector to:

- Track visibility state per observer
- Perform targeted visibility validation
- Avoid unnecessary calculations

Layer Mask Configuration

Obstacle Layers

Obstacle layers determine which objects block visibility.

Examples:

Walls
Buildings
Terrain
Static Geometry

Only layers included in the obstacle mask participate in line-of-sight validation.

Incorrect layer configuration is one of the most common causes of visibility issues.

Debug Visualization

PlayerVisibilityDetector provides extensive runtime visualization tools.

Available debug views include:

Fixed Lines
Dynamic Lines
Detected Lines
Blocked Lines
Visibility
Volume Face
Normals

These visualizations help diagnose:

- Visibility issues
 - Incorrect collider sizing
 - Sampling density problems
 - Layer mask configuration errors
-

Performance Recommendations

For most projects:

Moderate Fixed Lines
Moderate Dynamic Lines
32 Updates Per Second
Velocity Compensation
Enabled

provides an excellent balance between responsiveness and performance.

For large multiplayer environments:

- Reduce sampling density
 - Lower update frequency
 - Reduce view distance
 - Profile before increasing line counts
-

Common Mistakes

Excessive Sampling Density

Increasing every sampling setting simultaneously can significantly increase CPU usage without noticeable gameplay improvements.

Incorrect PlayerBox Size

Boxes that are too large:

Early Detection

Boxes that are too small:

Delayed Detection

Incorrect Layer Masks

If obstacle layers are not configured correctly:

False Positives
False Negatives
Visibility Inconsistencies

may occur.

Best Practices

- Start with moderate sampling values
- Use velocity compensation for multiplayer games
- Keep obstacle masks clean and organized
- Profile performance before increasing line density
- Use debug visualization during integration

PlayerVisibilityDetector is the component responsible for performing the final visibility validation and is one of the most important parts of the Anti-Wallhack Visibility System.

6. PlayerBox

Overview

PlayerBox defines the visibility volume used by the Anti-Wallhack Visibility System.

Unlike traditional gameplay colliders, PlayerBox is not intended to represent physical collision. Instead, it defines the area used by the visibility system when determining whether a player should be considered visible.

All visibility sampling performed by PlayerVisibilityDetector is based on the dimensions provided by PlayerBox.

Proper PlayerBox configuration is one of the most important factors affecting visibility accuracy.

Purpose

PlayerBox serves as the geometric representation of a visibility target.

The component is responsible for:

- Defining the visibility volume
- Providing sampling boundaries
- Supporting velocity compensation
- Generating visibility points
- Improving visibility consistency

The visibility system uses this volume to determine where visibility samples should be placed.

Configuration

Center

Center

Defines the local offset of the visibility volume.

This allows the visibility volume to be aligned with the visual representation of the character.

Examples:

- Aligning the box with the torso
- Adjusting for custom character pivots
- Correcting imported model offsets

Size

Size

Defines the base dimensions of the visibility volume.

This represents the standard visibility area when no compensation is applied.

Typical examples:

FPS Character

Width: 0.6
Height: 1.8
Depth: 0.6

TPS Character

Width: 0.8
Height: 1.8
Depth: 0.8

Actual values should be adjusted according to the visual size of the character.

Expanded Size

Expanded Size

Defines the maximum expansion applied to the visibility volume.

This value is used by the velocity compensation system.

As movement speed increases, the visibility volume can temporarily expand toward the configured limits.

Benefits:

- Improved fairness
 - Better corner peeking behavior
 - Reduced visibility inconsistencies
-

Velocity Compensation Integration

The Free Edition includes velocity-based expansion.

When enabled, the system adjusts the visibility volume according to player movement speed.

Example:

Standing Still

↓

Base Size

Running

↓

Expanded Size

This allows fast-moving targets to become visible more consistently during rapid movement.

Visibility Accuracy

PlayerBox directly affects visibility behavior.

Oversized Boxes

If the visibility volume is significantly larger than the visual character:

Earlier Detection
Increased Visibility Area
Reduced Precision

This may cause players to become visible sooner than expected.

Undersized Boxes

If the visibility volume is significantly smaller than the visual character:

Delayed Detection
Reduced Visibility Area
Potential Exposure
Issues

This may allow players to expose parts of their character before visibility is properly detected.

Recommended Sizing

A good PlayerBox should:

- Cover the main body of the character
- Match the visual silhouette
- Exclude unnecessary empty space
- Remain consistent across all player characters

Avoid using extremely aggressive dimensions.

Moderate sizing typically produces the most natural results.

Character Variations

Different character types may require different PlayerBox configurations.

Examples:

Standard Human Character

Medium Width
Standard Height

Heavy Character

Increased Width
Standard Height

Small Character

Reduced Width
Reduced Height

The visibility volume should always reflect the intended gameplay balance.

Debug Visualization

PlayerBox integrates with the visibility debugging system.

When gizmos are enabled, developers can visualize:

Visibility Volume
Expanded Volume
Sampling Area
Detection Boundaries

This allows rapid adjustment and validation during development.

Performance Considerations

PlayerBox itself has a negligible performance cost.

However, larger visibility volumes may:

- Increase visibility candidates
- Increase successful visibility checks
- Increase network synchronization frequency

For best results:

- Use realistic dimensions
 - Avoid excessive expansion values
 - Profile changes in multiplayer environments
-

Best Practices

- Match the box to the visible body shape
- Use moderate expansion values
- Validate sizing with debug gizmos
- Test corner peeking scenarios
- Keep dimensions consistent across player characters

PlayerBox serves as the foundation of the Free Edition visibility system and plays a major role in determining the accuracy and fairness of visibility validation.

7. Network Adapter Components

Overview

The Free Edition includes default networking adapter components designed to simplify integration between the Anti-Wallhack Visibility System and multiplayer frameworks.

These components act as a bridge between the visibility system and the networking solution used by the project.

Their purpose is to initialize the visibility system and forward update execution to the core visibility components.

The included adapters are intended as reference implementations and can be customized to fit the requirements of a specific networking framework.

Why Network Adapters Exist

The core visibility system is framework-agnostic.

Components such as:

ObserverManager
PlayerObserver
PlayerVisibilityDetector
PlayerBox

do not depend on any specific networking solution.

Instead, the network adapters are responsible for connecting those components to the update loop used by the multiplayer framework.

This design allows the visibility system to remain flexible and compatible with multiple networking solutions.

NetworkPlayerObserver

Purpose

NetworkPlayerObserver acts as the integration layer for PlayerObserver.

It initializes the observer and forwards execution calls to the visibility system.

The default implementation is intentionally simple and can be adapted to any networking framework.

Responsibilities

NetworkPlayerObserver is responsible for:

- Initializing PlayerObserver
 - Registering visibility event callbacks
 - Forwarding observer updates
 - Maintaining observer visibility state
 - Providing a starting point for networking integration
-

Initialization

During startup:

```
Awake()  
↓  
InitializeSystem()  
↓  
Observer Registration
```

The adapter automatically initializes the observer and prepares it for visibility validation.

Server Tick Execution

The adapter forwards update execution to:

```
PlayerObserver.ServerTick(time)
```

Example:

```
private void Update()  
{  
    var time = Time.time;  
    playerObserver.ServerTick(time);  
}
```

The included implementation uses Unity's Update method for demonstration purposes.

Visibility Events

NetworkPlayerObserver automatically subscribes to:

ObserverManager.OnVisibilityChanged

and updates:

playerObserver.IsVisible

whenever the visibility state changes.

This allows visibility information to be consumed by rendering systems, gameplay systems, or networking logic.

NetworkPlayerVisibilityDetector

Purpose

NetworkPlayerVisibilityDetector acts as the integration layer for PlayerVisibilityDetector.

It initializes the detector and forwards visibility execution calls.

Responsibilities

NetworkPlayerVisibilityDetector is responsible for:

- Initializing PlayerVisibilityDetector
- Forwarding visibility updates
- Providing a networking integration example
- Connecting the detector to the game update loop

Initialization

During startup:

Awake()

↓

InitializeSystem()

↓

Player Registration

The detector is automatically prepared for visibility validation.

Server Tick Execution

The adapter forwards execution to:

```
PlayerVisibilityDetector.ServerTick(time)
```

Example:

```
private void Update()  
{  
    var time = Time.time;  
    playerVisibilityDetector.ServerTick(time);  
}
```

The included implementation is intended as a starting point and should be adapted for multiplayer environments.

Adapting to Networking Frameworks

The default adapters use Unity's Update method because they must remain framework-independent.

However, multiplayer projects should execute visibility validation using the networking framework's server-authoritative update loop.

Examples:

```
Photon Fusion  
Mirror  
Fish-Networking  
Netcode for  
GameObjects Photon  
PUN
```

The exact implementation depends on the framework being used.

Additional integration examples are available in the online documentation.

Server-Authoritative Execution

For multiplayer games, visibility validation should always execute on the server or host authority.

Recommended flow:

Network Tick

↓

ServerTick()

↓

Visibility Validation

↓

Visibility Result

↓

Network Synchronization

This ensures that visibility decisions remain authoritative and resistant to client-side manipulation.

Important Update Notice

The included adapter components are intended as reference implementations.

Most projects will eventually customize them to match their networking architecture.

After adapting the components to your project, it is recommended to maintain your own customized versions.

Reason:

Future asset updates may replace the default adapter scripts.

If the adapters have been heavily modified, importing a newer version of the asset may overwrite those customizations.

Recommended Workflow

1. **Import the asset**
2. **Configure the default adapters**
3. **Adapt them to your networking framework**
4. **Move customized versions into your project architecture**
5. **Keep your project-specific implementation under source control**

This approach minimizes maintenance effort and prevents accidental loss of integration code.

Best Practices

- Execute visibility validation on the server or host
- Use the networking framework's fixed simulation tick whenever possible

- Keep adapters lightweight
- Avoid placing gameplay logic inside the adapters
- Treat the adapters as integration layers rather than gameplay systems
- Maintain customized versions after integration

The Network Adapter Components provide a simple and flexible foundation for connecting the Anti-Wallhack Visibility System to virtually any Unity multiplayer framework.

8. Visibility Pipeline

Overview

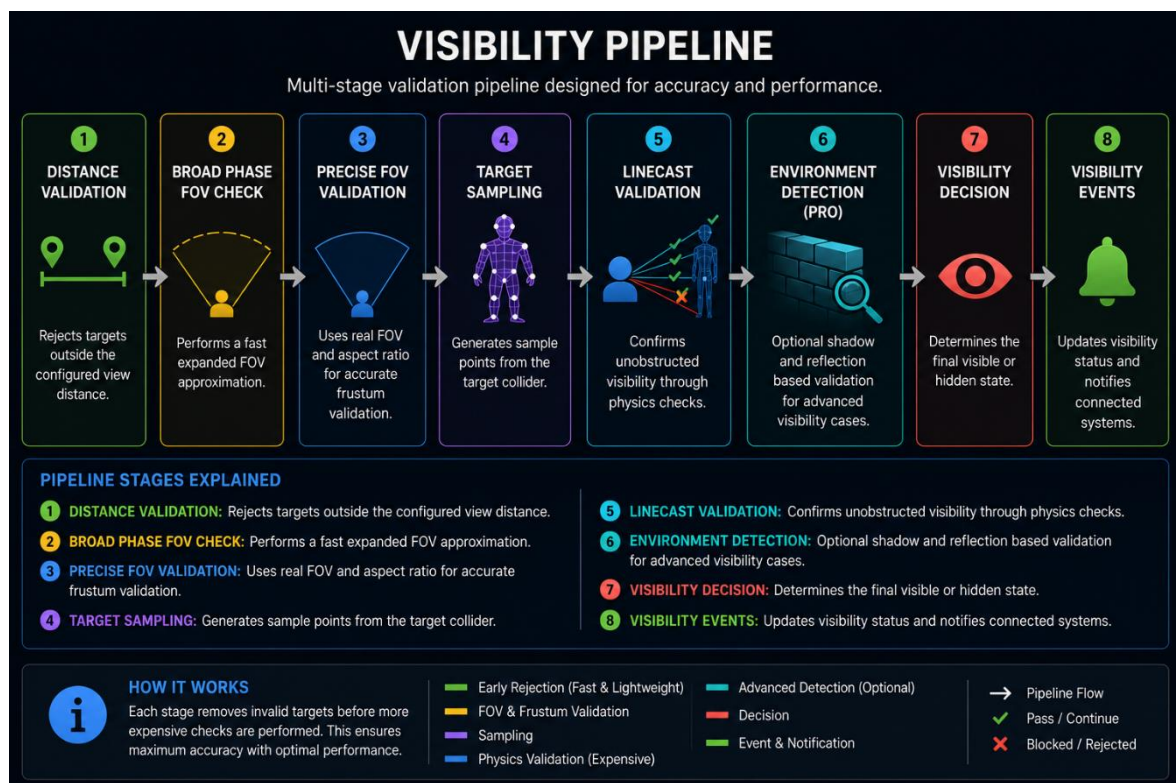
The Anti-Wallhack Visibility System uses a multi-stage visibility pipeline designed to maximize performance while maintaining accurate visibility validation.

Instead of immediately performing expensive line-of-sight calculations against every target, the system progressively filters candidates through multiple validation stages.

This approach allows large numbers of players to be processed efficiently while maintaining reliable visibility results.

The visibility pipeline is one of the most important concepts in the system and understanding it helps developers tune performance and accuracy more effectively.

Visibility Pipeline



Broad Phase

↓

Distance Validation

↓

FOV Validation

↓

Visibility Sampling

↓

Final Visibility Result

Each stage progressively reduces the number of candidates that require further processing.

Stage 1 – Broad Phase

Purpose

The Broad Phase is the first optimization layer.

Its goal is to quickly eliminate targets that are obviously outside the observer's area of interest.

Before any detailed calculations occur, the system evaluates whether a target falls within an expanded detection region.

This stage uses:

broadPhaseFOVMultiplier

Benefits:

- Very low computational cost
- Quickly rejects invalid targets
- Reduces workload for later stages

Without Broad Phase filtering, every target would proceed to expensive visibility calculations.

Stage 2 – Distance Validation

Purpose

The second stage validates distance.

Targets beyond the observer's configured visibility range are immediately discarded.

The system compares:

Distance To Target

against:

viewDistance

Benefits:

- Extremely fast calculation
- Large reduction in visibility candidates
- Important for large multiplayer maps

Distance validation is one of the most effective performance optimizations available.

Stage 3 – FOV Validation

Purpose

Targets that pass Broad Phase and Distance validation proceed to Field of View validation.

At this stage the system evaluates:

Observer Direction

against:

Target Position

using:

cameraFOVAngle
aspectFormat
maxObserverAngle

Only targets located inside the observer's viewing region continue through the pipeline.

180° Observer

In 180° mode:

Targets Behind The Observer

are rejected immediately.

This significantly reduces visibility workload and is recommended for most multiplayer games.

360° Observer

In 360° mode:

All Directions

are considered valid.

This increases flexibility but may increase visibility processing costs.

Stage 4 – Visibility Sampling

Purpose

After passing all previous stages, the target proceeds to visibility sampling.

This is where the actual visibility validation occurs.

The system generates visibility samples using:

Fixed Lines
Dynamic Lines

generated from:

PlayerBox

The detector then performs physics-based visibility checks.

Fixed Lines

Provide stable visibility coverage.

Benefits:

- Consistent results
 - Predictable behavior
 - Low overhead
-

Dynamic Lines

Provide moving visibility samples.

Benefits:

- Better visibility coverage
 - Improved corner detection
 - Improved detection through small openings
-

Velocity Compensation

If enabled, the visibility volume may expand according to target movement speed.

Benefits:

- Fairer visibility behavior
 - Better handling of fast-moving targets
 - Reduced corner-peeking inconsistencies
-

Stage 5 – Final Visibility Result

Purpose

The final stage determines whether the target should be considered visible.

If visibility sampling succeeds:

Visible

is returned.

Otherwise:

Hidden

is returned.

The visibility state is then sent to:

ObserverManager

which updates visibility information and dispatches visibility events when necessary.

Why The Pipeline Exists

A naive implementation would perform line-of-sight checks against every target.

Example:

100 Players

↓

100 Visibility Calculations

The Anti-Wallhack Visibility System instead performs progressive filtering:

100 Players

↓

Broad Phase

↓

35 Players

↓

Distance Validation

↓

18 Players

↓

FOV Validation

↓

7 Players

↓

Visibility Sampling

This dramatically reduces the number of expensive physics checks.

Performance Benefits

The visibility pipeline provides:

- Better scalability
- Lower CPU usage
- More predictable performance
- Improved multiplayer support

Most visibility candidates are eliminated before visibility sampling is ever performed.

This is one of the primary reasons the system remains efficient even in multiplayer environments with large numbers of players.

Best Practices

- Use 180° observer mode whenever possible
- Tune view distance according to map size
- Keep Broad Phase enabled
- Avoid excessive visibility ranges
- Use appropriate sampling density
- Profile performance before increasing accuracy settings

The Visibility Pipeline forms the foundation of the Anti-Wallhack Visibility System and is responsible for balancing accuracy, fairness, and performance.

9. Visibility Sampling System

Overview

The Visibility Sampling System is the core mechanism responsible for determining whether a target should be considered visible.

After a target passes the observer validation stages, the system performs a series of visibility samples distributed across the target's visibility volume.

Unlike simple raycast-based approaches that rely on a single line-of-sight test, the Anti-Wallhack Visibility System uses multiple samples to produce more reliable visibility results.

This approach reduces false negatives while maintaining scalability.

Why Sampling Is Necessary

A single raycast can produce unreliable visibility results.

Examples:

Small Openings Corners Railings Windows Partial Exposure

A single visibility point may incorrectly classify a target as hidden even when part of the target is actually visible.

The Visibility Sampling System solves this problem by evaluating multiple points across the visibility volume.

Sampling Sources

The Free Edition uses two sampling systems:

Fixed Lines

Dynamic Lines

These systems complement each other and work together to improve visibility accuracy.

Fixed Lines

Purpose

Fixed Lines provide permanent visibility samples distributed across the visibility volume.

They represent the baseline visibility coverage of the system.

Fixed lines never move and remain stable throughout gameplay.

Advantages

Fixed Lines provide:

- Consistent visibility checks
- Predictable behavior
- Stable visibility validation
- Low computational overhead

Because they remain static, they are highly efficient and easy to debug.

Configuration

The number of fixed samples is controlled by:

`verticalFixedLineCount` `horizontalFixedLineCount`

Increasing these values:

More Samples

↓

Higher Accuracy

↓

Higher CPU Cost

Reducing these values:

Fewer Samples ↓ Lower CPU Cost ↓ Lower Accuracy

Dynamic Lines

Purpose

Dynamic Lines continuously move across the visibility volume.

Their objective is to sample regions that Fixed Lines may not permanently cover.

This allows the system to detect visibility opportunities that occur between fixed samples.

Advantages Dynamic

Lines provide:

- **Better coverage over time**
- **Improved corner visibility detection**
- **Improved detection through narrow openings**
- **Increased sampling flexibility**

Dynamic Movement

Unlike Fixed Lines:

Static Position

Dynamic Lines:

Move Across The Visibility Volume

This movement gradually scans additional visibility regions without permanently increasing sample density.

Configuration

Dynamic behavior is controlled by:

verticalDynamicLineCount
horizontalDynamicLineCount
dynamicLineSpacing
dynamicLineSpeed

Dynamic Line Speed

dynamicLineSpeed

dynamicLineSpeed

Defines how quickly Dynamic Lines move across the visibility volume.

Higher values:

- **Faster visibility coverage**
- **Improved responsiveness**
- **More frequent sampling**

changes Lower values:

- **Slower coverage**
- **Reduced processing variation**
- **More stable sampling patterns**

Recommended Values

Most projects perform well using:

5 - 15

Developers should profile and adjust according to gameplay requirements.

Dynamic Line Spacing

dynamicLineSpacing

dynamicLineSpacing

Controls the distance between Dynamic Lines.

Smaller spacing:

More Coverage
Higher Accuracy
Higher Cost

Larger spacing:

Less Coverage Lower Cost

Grid Subdivision

The visibility volume is divided into a configurable grid. This grid determines where visibility samples are generated. The subdivision is controlled by:

**horizontalColumnCount
verticalRowCount**

Higher Density

Higher subdivision values produce:

- More visibility samples
- Better visibility precision
- Increased CPU usage

Recommended for:

Competitive Multiplayer Games

Lower Density

Lower subdivision values produce:

- **Fewer visibility samples**
- **Reduced CPU usage**
- **Faster**

processing

Recommended for:

Large-Scale Multiplayer Games

Sampling Accuracy

The overall accuracy of the visibility system depends on the combination of:

Fixed Lines Dynamic Lines Grid Density PlayerBox Size

Increasing every setting simultaneously is rarely necessary. A balanced configuration usually provides the best results.

Interaction With Velocity Compensation

Velocity Compensation works together with visibility sampling.

When movement speed increases:

Player Speed ↑

↓

Visibility Volume Expansion ↑

↓

Sampling Coverage ↑

This allows visibility checks to adapt naturally to movement.

Debug Visualization

The Visibility Sampling System includes extensive debugging support.

Available debug views include:

Fixed Lines Dynamic Lines Detected Lines Blocked Lines Visibility Volume

Debug visualization is strongly recommended during initial integration and tuning.

Performance vs Accuracy

One of the most important tuning decisions is balancing accuracy against processing cost.

Accuracy-Oriented Configuration

Example:

High Fixed Line Count
High Dynamic Line Count
Dense Grid

Results:

- Better visibility precision
- Better corner detection
- Higher CPU cost

Performance-Oriented Configuration

Example:

Lower Fixed Line Count Lower Dynamic Line Count Reduced Grid Density

Results:

- Lower CPU usage
- **Better scalability**
- **Slightly reduced precision**

Recommended Starting Configuration

For most multiplayer games:

Moderate Fixed Lines
Moderate Dynamic Lines
32 Updates Per Second
Velocity Compensation
Enabled

provides an excellent balance between responsiveness and performance.

Developers should start with conservative values and increase density only when necessary.

Best Practices

- Begin with moderate sampling density
- Profile before increasing line counts
- Use Dynamic Lines to improve coverage instead of excessive Fixed Lines
- Validate changes using debug gizmos
- Test visibility around corners and small openings
- Balance performance and accuracy according to game scale

The Visibility Sampling System is the final stage responsible for determining visibility and plays a central role in the accuracy and fairness of the Anti-Wallhack Visibility System.

10. Visibility Filtering

Overview

Many multiplayer games require certain entities to remain synchronized regardless of visibility.

Examples include:

Teammates
Friendly
NPCs Squad
Members Pets
Minimap Targets
Revive Targets

In these situations, visibility validation should not determine whether those entities are synchronized to the client.

The planned Visibility Filtering system will allow developers to selectively exclude targets from visibility validation while still benefiting from the Anti-Wallhack Visibility System for all other entities.

Why Visibility Filtering Is Important

Consider a team-based game.

Example:

Player A
↓
Blue Team

Player B
↓
Blue Team

If Player B moves behind a wall:

Wall
↓
Player B

the current visibility system may determine:

Player B = Hidden

and synchronization may stop.

However, many games still need to know the location of teammates for:

- Squad indicators
- Team markers
- Nameplates
- Revive systems
- Minimap displays
- Tactical communication

In these cases, teammates should bypass visibility validation entirely.

Filtering System

The planned Visibility Filtering system will allow developers to decide whether a target should participate in visibility validation.

Conceptually:

Target

↓

Visibility Filter

↓

Validate Visibility?

↓

Yes / No

If validation is skipped, the target remains synchronized normally.

Example Use Cases

Team-Based Games

Allow teammates to remain synchronized at all times.

Example:

Blue Team

Always Visible To Blue Team

while enemy players continue using visibility validation.

Friendly NPCs

Friendly AI characters may always remain visible to the player.

Example:

Friendly NPC

↓

Ignore Visibility Validation

Pets And Companions

Companion entities may remain synchronized regardless of visibility state.

Example:

Pet

↓

Always Visible

Spectator Systems

Spectator cameras may require unrestricted visibility.

Example:

Spectator

↓

Ignore Visibility Restrictions

API Design

The exact implementation may evolve over time.

One possible approach is a visibility filter callback:

```
bool ShouldValidate(  
    PlayerObserver observer,  
    PlayerVisibilityDetector target);
```

The callback would determine whether visibility validation should proceed.

Example

ShouldValidate()

↓

True

↓

Run Visibility Validation

Skipped Validation

ShouldValidate()

↓

False

↓

Keep Target Synchronized

Benefits

Visibility Filtering provides:

- Better support for team-based games
- Easier integration with existing gameplay systems
- Reduced custom networking code
- Improved flexibility
- Better compatibility with minimap and squad systems

Performance Impact

Visibility Filtering may slightly improve performance because filtered targets can bypass expensive visibility calculations.

Example:

Teammates

↓

Skip Validation

↓

Reduced Visibility Workload

The impact depends on the number of filtered entities.

Recommended Use Cases

Visibility Filtering is particularly useful for:

**Competitive Team
Shooters Battle Royale
Games Extraction
Shooters
Co-op Games
Tactical Games
Squad-Based**

Future Development

Visibility Filtering is planned as a future Free Edition feature.

The goal is to provide a lightweight and framework-agnostic solution that can be integrated into any multiplayer architecture.

The final implementation may include:

- Team filtering
- Friendly NPC filtering
- Custom entity filtering
- User-defined visibility rules

Additional details and implementation examples will be published in the online documentation as the feature evolves.

Best Practices

When available:

- Filter teammates whenever gameplay requires permanent team awareness
- Continue validating enemy players through the visibility system
- Keep filtering rules simple and predictable
- Avoid excessive custom filtering logic

Visibility Filtering is intended to improve flexibility while maintaining the security benefits of server-authoritative visibility validation.

11. Networking Integration

Overview

The Anti-Wallhack Visibility System is designed to be networking-framework agnostic.

The visibility system itself does not depend on any specific multiplayer solution and can be integrated into a wide range of server-authoritative architectures.

This approach allows developers to use the visibility system with their preferred networking framework while maintaining a consistent visibility validation workflow.

Supported Frameworks

The system has been designed to integrate with:

Photon Fusion
Photon PUN 2
Mirror
Fish-Networking
Netcode for GameObjects (NGO)

Additional frameworks can also be supported through custom integration adapters.

Integration Philosophy

The visibility system should not control the networking framework.

Instead, the networking framework should control when visibility updates are executed.

Recommended architecture:

Networking Framework
↓
Network Adapter Components
↓
ServerTick()
↓
Visibility Validation
↓
Visibility Result
↓
Network Synchronization

This keeps the visibility system independent from networking-specific logic.

Server-Authoritative Execution

For maximum security, visibility validation should always execute on the authoritative side of the game.

Depending on the architecture, this may be:

Dedicated Server
Host
Server Authority Instance

Visibility calculations should never rely on client-side authority.

Synchronization Workflow

A typical visibility workflow looks like:

Observer

↓

Visibility Validation

↓

Visible?

↓

Yes / No

↓

Network Synchronization

Only entities considered visible should be synchronized to the observing player.

This reduces unnecessary network traffic and helps prevent information leaks that could be exploited by wallhack tools.

Framework Integration Examples

The Free Edition includes generic adapter components:

NetworkPlayerObserver
NetworkPlayerVisibilityDetect
or

These components provide a simple starting point and demonstrate the expected integration pattern.

The included implementations use Unity's Update loop for simplicity.

Production multiplayer projects should move visibility execution to the framework's server tick.

Photon Fusion

Photon Fusion is one of the recommended networking solutions for competitive multiplayer games.

Typical integration:

FixedUpdateNetwork()

↓

ServerTick()

Recommended authority:

State Authority

Visibility validation should only execute on the authoritative simulation.

Mirror

Mirror integrations typically execute visibility updates inside the server update cycle.

Recommended flow:

```
graph TD;
  A[Server] --> B[ServerTick()];
  B --> C[Visibility Validation];
```

Clients should receive only the final visibility result.

Fish-Networking

Fish-Networking follows a similar server-authoritative model.

Visibility validation should execute inside the framework's network tick.

Recommended flow:

```
graph TD;
  A[Server Tick] --> B[ServerTick()];
  B --> C[Visibility Result];
```

Netcode for GameObjects

Netcode for GameObjects supports server-authoritative architectures and can integrate directly with the visibility system.

Recommended flow:

```
graph TD;
  A[Server] --> B[Visibility Validation];
  B --> C[RPC / Synchronization];
```

The visibility system should execute before replication decisions are made.

Photon PUN 2

Photon PUN projects often use a host-authoritative architecture.

In these cases, visibility validation should execute only on the authoritative host.

Recommended flow:

Master Client

↓

Visibility Validation

↓

Synchronization

Online Integration Examples

Because networking frameworks evolve frequently, complete integration examples are maintained in the online documentation.

The online documentation contains:

- Updated integration examples
- Framework-specific setup instructions
- Best practices
- Troubleshooting information
- Future framework support

The online documentation should always be considered the primary source for networking integration guidance.

Common Integration Mistakes

Executing Visibility Validation On Every Client

Incorrect:

Client A

Client B

Client C

↓

Visibility Validation

This creates inconsistent visibility states and weakens security.

Correct:

```
Server
↓
Visibility Validation
↓
Clients
```

Running Visibility Validation Before Initialization

Incorrect:

```
ServerTick()
↓
InitializeSystem()
```

Correct:

```
InitializeSystem()
↓
ServerTick()
```

All visibility components must be initialized before updates begin.

Mixing Gameplay Logic With Adapter Components

The adapter components should remain lightweight.

Avoid placing:

- Gameplay rules
- Combat systems
- Inventory systems
- Team management systems

inside the network adapters.

Their primary responsibility is visibility integration.

Best Practices

- Execute visibility validation only on the authoritative side
- Use the framework's fixed network tick whenever possible
- Keep adapters lightweight
- Separate visibility logic from gameplay logic
- Follow framework-specific synchronization recommendations
- Refer to the online documentation for updated integration examples

The networking integration layer allows the Anti-Wallhack Visibility System to remain flexible, scalable, and compatible with a wide range of multiplayer architectures.

12. Velocity Compensation

Overview

Velocity Compensation is a feature included in the Free Edition that improves visibility validation for moving targets.

In traditional visibility systems, visibility calculations are often performed using a fixed visibility volume regardless of movement speed.

This can create situations where a fast-moving player becomes partially exposed before the visibility system properly detects them.

Velocity Compensation addresses this issue by dynamically expanding the visibility volume based on movement speed.

The result is a more consistent and fair visibility experience, especially in fast-paced multiplayer games.

Why Velocity Compensation Exists

Consider the following situation:

Player Running
↓
Approaches Corner
↓
Partially Exposed
↓
Visibility Check

Without compensation, a fast-moving player may visually expose themselves before their visibility volume is sampled effectively.

This can create situations where:

- The moving player can see first
- Detection occurs later than expected
- Visibility feels inconsistent

Velocity Compensation reduces these issues by anticipating movement exposure.

Corner Peeking

One of the most common visibility problems in multiplayer games occurs during corner peeking.

Example:

Wall



Player A

Running

As Player A moves around the corner, part of the character becomes visible before the center of the visibility volume fully exits cover.

Without compensation:

Visual Exposure



Visibility Detection

With Velocity Compensation:

Movement Prediction



Expanded Visibility Volume



Earlier Detection

This creates more natural visibility behavior.

How It Works

Velocity Compensation expands the visibility volume according to movement speed.

Conceptually:

Movement Speed ↑



Visibility Volume Expansion ↑

When a player is standing still:

Base Size

is used.

When movement speed increases:

Expanded Size

gradually becomes active.

This expansion allows visibility samples to cover the player's expected exposure area more effectively.

Expanded Size

Velocity Compensation relies on the PlayerBox configuration.

Two dimensions are used:

Size
Expanded
Size

Size

Represents the default visibility volume.

Example:

Standing Player

Expanded Size

Represents the maximum visibility volume that may be used during movement.

Example:

Running Player

The system automatically interpolates between both volumes according to movement speed.

Benefits

Velocity Compensation provides several gameplay advantages:

Fairer Visibility

Players become visible closer to the moment they visually expose themselves.

Improved Corner Behavior

Visibility transitions become smoother when moving around obstacles.

Better Multiplayer Experience

High-speed movement becomes more predictable for both players.

Reduced Visibility Inconsistencies

The visibility system reacts more naturally to rapid movement.

Performance Impact

Velocity Compensation has a very low performance cost.

The system does not perform additional raycasts or visibility passes.

Instead, it adjusts the visibility volume already used by the sampling system.

Benefits:

Low CPU Cost
Low Memory
Cost
High Gameplay Benefit

Because of this, Velocity Compensation is recommended for most multiplayer games.

Configuration Recommendations

Conservative Configuration

Recommended for:

Tactical Shooters
Competitive Games

Characteristics:

- Small expansion values
 - Higher precision
 - Lower visibility permissiveness
-

Balanced Configuration

Recommended for:

General Multiplayer Games

Characteristics:

- Moderate expansion values
 - Good balance between fairness and precision
-

Aggressive Configuration

Recommended for:

Fast-Paced Arcade Games

Characteristics:

- Larger expansion values
 - Earlier visibility detection
 - More forgiving movement behavior
-

Common Mistakes

Excessive Expansion

Very large expansion values may cause players to become visible too early.

Example:

Player Still Behind Cover

↓

Visibility Triggered

This can make visibility feel overly aggressive.

No Expansion

Disabling or minimizing expansion may reintroduce corner-peeking inconsistencies.

Example:

Player Exposed

↓

Visibility Triggered Too Late

Oversized PlayerBox

Combining large PlayerBox dimensions with aggressive expansion values can lead to excessive visibility.

Always validate sizing using debug visualization.

Testing Velocity Compensation

The recommended testing procedure is:

1. **Create a wall**
2. **Create two players**
3. **Move around corners at different speeds**
4. **Observe visibility transitions**
5. **Adjust Expanded Size if necessary**

Debug visualization should be enabled during tuning.

Using Velocity Compensation As Latency Mitigation

The Free Edition does not include dedicated latency-aware visibility compensation systems.

As a result, players with higher latency may occasionally experience visibility transitions that occur slightly later than expected during fast movement or corner peeking situations.

To partially mitigate this behavior, developers can increase the velocity compensation settings.

Conceptually:

Higher Ping

↓

Higher Movement Uncertainty

↓

Increased Velocity Compensation

↓

Earlier Visibility Detection

This approach does not replace true latency compensation, but it can help reduce visibility inconsistencies in multiplayer environments where moderate network latency is expected.

Recommended Usage

Scenario	Recommendation
LAN / Local Network	Conservative Velocity Compensation
Typical Online Multiplayer	Balanced Velocity Compensation
High-Ping Environments	More Aggressive Velocity Compensation

Important Considerations

Increasing velocity compensation may cause players to become visible slightly earlier than their actual visual exposure.

Developers should balance:

**Fairness
vs
Visibility Precision**

to find the configuration that best matches their game's networking conditions.

Professional Edition

The Professional Edition introduces dedicated latency-aware visibility systems that use network latency information directly during visibility calculations.

For this reason, aggressive velocity compensation should be considered a practical workaround rather than a replacement for true latency compensation.

Best Practices

- Enable Velocity Compensation for multiplayer games
- Start with moderate expansion values
- Test corner-peeking situations frequently
- Use realistic PlayerBox dimensions
- Avoid excessive expansion values
- Increase compensation gradually when supporting higher latency environments
- Validate changes using debug gizmos

Summary

Velocity Compensation is one of the most important features included in the Free Edition.

It improves fairness during movement, reduces corner-peeking inconsistencies, and can even be used as a practical latency-mitigation technique in environments where dedicated latency-aware visibility systems are not available.

When properly configured, it provides a significant gameplay benefit while maintaining a very low performance cost.

13. Performance Optimization

Overview

The Anti-Wallhack Visibility System has been designed to scale efficiently by combining multiple filtering stages, configurable update rates, and optimized visibility sampling.

While the system is highly configurable, proper tuning is essential to achieve the best balance between visibility accuracy and server performance.

This chapter explains the most important settings that affect performance and provides practical recommendations based on benchmark testing.

Understanding Visibility Cost

The total workload of the visibility system depends on several factors:

Number of Players

↓

Number of Observers

↓

Visibility Candidates

↓

Visibility Sampling

↓

CPU Usage

In general, performance is affected more by the number of visibility candidates than by the total number of players present in the game.

A player located on the opposite side of a large map often generates no meaningful visibility workload because they are rejected early by the pipeline.

Observer Update Rate

One of the most important performance settings is the ObserverManager update rate.

Recommended Configuration

32 Updates Per Second

This configuration was used during the official benchmark tests and provides an excellent balance between responsiveness and performance.

Benefits:

- Responsive visibility updates
 - Stable CPU usage
 - Suitable for most multiplayer games
-

Lower Update Rates

Examples:

16 TPS
20 TPS

Benefits:

- Lower CPU usage
- Better scalability
- Suitable for large player counts

Trade-off:

- Slightly slower visibility updates
-

Higher Update Rates

Examples:

48 TPS
64 TPS

Benefits:

- Faster visibility updates
- Improved responsiveness

Trade-off:

- Increased CPU usage

Higher update rates should only be used when gameplay requirements justify the additional cost.

180° vs 360° Observer Mode

Observer angle selection has a significant impact on performance.

180° Mode

Recommended

The observer only evaluates targets located within the viewing direction.

Benefits:

- Fewer visibility candidates
- Lower CPU usage
- Better scalability

Recommended for:

- FPS games
 - TPS games
 - Tactical shooters
 - Survival games
-

360° Mode

The observer evaluates targets in every direction.

Benefits:

- Omnidirectional awareness
- Useful for AI systems
- Useful for top-down games

Trade-offs:

- More visibility candidates
- Increased CPU usage

The official benchmark tests demonstrated that 360° mode generates a noticeably higher workload than 180° mode because more players pass the observer filtering stages.

View Distance

View distance directly affects how many targets reach visibility validation.

Large View Distance

Benefits:

- Longer detection range

Trade-offs:

- More visibility candidates
 - Increased CPU usage
-

Small View Distance

Benefits:

- Better scalability
- Fewer visibility calculations

Recommended for:

- Indoor maps
 - Arena shooters
 - Small multiplayer environments
-

Detection Modes

PlayerObserver provides two detection modes.

ClosestPoint

Performance-Oriented

Benefits:

- Lowest CPU cost
- Fast visibility evaluation

Recommended for:

- Large player counts
 - Performance-sensitive environments
-

BoundsCorners

Accuracy-Oriented

Benefits:

- Improved edge visibility handling
- Better precision

Trade-offs:

- More calculations

Recommended for:

- Competitive multiplayer games
- Tactical shooters
- High-precision environments

Visibility Sampling Density

Sampling density has a direct impact on visibility workload.

The main contributors are:

Fixed Lines
Dynamic
Lines Grid
Density

High Density

Benefits:

- Better visibility precision
- Better corner detection
- Improved small opening detection

Trade-offs:

- Increased CPU usage

Moderate Density

Recommended for most games.

Provides:

- Good visibility quality
 - Good scalability
 - Balanced performance
-

Low Density

Recommended for:

- Large-scale games
 - Very high player counts
 - Dedicated servers with strict performance budgets
-

Velocity Compensation

Velocity Compensation has a very small performance impact.

Because it primarily adjusts visibility volume dimensions rather than introducing additional visibility passes, it is generally recommended to keep it enabled.

Benefits:

- Better corner behavior
 - Improved fairness
 - Minimal CPU cost
-

Debug Visualization

Debug visualization should be used during development and testing.

However, excessive debug rendering may affect editor performance.

Recommended workflow:

Development

↓

Debug Enabled

Production

↓

Debug Disabled

This ensures that benchmark and gameplay performance measurements remain representative.

Benchmark Configuration

The official benchmark tests were performed using:

**ObserverManager: 32 Updates Per
Second Observer Mode: 180°
Environment Detection: Disabled**

This configuration was selected because it represents a realistic production setup for most competitive multiplayer games.

Benchmark Results

SERVER-LIKE BENCHMARK RESULTS				
50m × 50m High-Density Stress Test • 2 Teams				
Players	FPS Impact	System Time	Cost / Player	Raycasts / Frame
10	1.44%	2.10 ms	0.210 ms	397
25	4.86%	5.72 ms	0.229 ms	949
50	13.44%	19.43 ms	0.389 ms	5,057
100	44.08%	61.46 ms	0.615 ms	15,008
Performance Summary				
Scenario		Result		
Typical Competitive Match (10 Players)		1.44% Impact		
Medium Match (25 Players)		4.86% Impact		
Large Match (50 Players)		13.44% Impact		
Stress Test (100 Players)		44.08% Impact		

These results represent a high-density benchmark environment designed to stress the visibility system.

Real multiplayer matches typically distribute players across larger areas, resulting in fewer simultaneous visibility candidates and lower average workload.

Scaling Recommendations

Small Games

Up to 20 Players

Recommended:

- 32 TPS
- BoundsCorners
- Moderate Sampling
- Velocity Compensation Enabled

Medium Games

20 - 50 Players

Recommended:

- 32 TPS
- Moderate Sampling
- Optimized View Distance

Large Games

50+ Players

Recommended:

- 16-32 TPS
- ClosestPoint
- Lower Sampling Density
- Conservative View Distance

General Optimization Strategy

When optimizing performance:

Adjust settings in this order:

1. **View Distance**
2. **Observer Mode**
3. **Update Rate**
4. **Detection Mode**
5. **Sampling Density**

This usually provides the largest performance gains with the smallest impact on visibility quality.

Best Practices

- Use 180° mode whenever possible
- Start with 32 TPS
- Keep view distance proportional to map size
- Use moderate sampling density
- Enable Velocity Compensation
- Profile before increasing accuracy settings
- Test under realistic multiplayer conditions

The Anti-Wallhack Visibility System is designed to provide scalable visibility validation while allowing developers to tune accuracy and performance according to the needs of their game.

14. Troubleshooting

Overview

This chapter covers the most common issues encountered when integrating the Anti-Wallhack Visibility System and provides recommended solutions.

Most problems are related to component setup, initialization order, layer masks, or networking integration.

Before investigating advanced issues, always verify that the system has been configured correctly.

Players Are Never Detected

Symptoms

Targets remain hidden at all times
No visibility events are triggered
No visible state changes occur

Possible Causes

ObserverManager Missing

Verify that an ObserverManager exists in the scene.

The visibility system requires a single active ObserverManager instance.

Components Not Initialized

Verify that:

PlayerObserver
PlayerVisibilityDetector

have been initialized correctly.

For networking integrations, ensure that:

InitializeSystem()

is executed before:

ServerTick()

View Distance Too Small

Verify that the target is located within:

viewDistance

configured in PlayerObserver.

Incorrect FOV Configuration

Verify:

cameraFOVAngle
aspectFormat
maxObserverAngle

Incorrect values may prevent targets from entering the visibility pipeline.

Players Become Visible Too Early

Symptoms

Targets visible before fully leaving cover
Visibility feels overly aggressive

Possible Causes

Oversized PlayerBox

Verify:

Size
Expanded
Size

Excessively large visibility volumes may increase detection range.

Excessive Velocity Compensation

Large expansion values may cause targets to become visible sooner than expected.

Reduce:

Expanded Size

and retest.

Players Become Visible Too Late

Symptoms

Players visually exposed before detection occurs
Corner peeking feels inconsistent

Possible Causes

Visibility Volume Too Small

Verify that PlayerBox dimensions match the visual size of the character.

Velocity Compensation Too Conservative

Increase velocity compensation gradually and retest movement scenarios.

Update Rate Too Low

Very low update rates may delay visibility updates.

Recommended starting value:

32 TPS

Visibility Flickering

Symptoms

Rapid Visible/Hidden transitions
Unstable visibility near walls

Possible Causes

State Stabilization Too Low

Increase:

stateStabilizeTime

to smooth transitions.

Very Small Visibility Openings

Windows, railings, and thin geometry may naturally produce visibility fluctuations.

Increase sampling density if necessary.

Visibility Validation Never Runs

Symptoms

No visibility updates
No visibility events

Possible Causes

ServerTick Not Executing

Verify that:

ServerTick()

is called regularly.

For multiplayer projects, ensure it executes inside the networking framework's authoritative update loop.

Components Disabled

Verify that:

NetworkPlayerObserver
NetworkPlayerVisibilityDetect
or

are enabled.

Debug Gizmos Are Not Visible

Symptoms

No lines visible in Scene View
No debugging information displayed

Possible Causes

Gizmos Disabled

Verify that Unity Gizmos are enabled in the Scene View.

Debug Options Disabled

Verify that the corresponding debug options are enabled in the component inspector.

Performance Is Lower Than Expected

Symptoms

High CPU usage
Unexpected visibility cost

Possible Causes

Excessive Sampling Density

Reduce:

Fixed Lines
Dynamic
Lines Grid
Density

Excessive View Distance

Reduce:

viewDistance

where possible.

Using 360° Mode Unnecessarily

Prefer:

180° Mode

whenever gameplay allows.

Update Rate Too High

Verify that ObserverManager is not running at an unnecessarily high update frequency.

Networking Synchronization Issues

Symptoms

Different visibility results across clients
Inconsistent visibility states

Possible Causes

Visibility Validation Executing On Clients

Visibility validation should execute only on the authoritative side.

Incorrect:

Client

↓

ServerTick()

Correct:

Server

↓

ServerTick()

Adapter Components Not Integrated Correctly

Review:

NetworkPlayerObserver
NetworkPlayerVisibilityDetector

and verify that they are connected to the networking framework's update loop.

Before Requesting Support

Before reporting an issue, verify:

- ✓ **ObserverManager exists**
- ✓ **Components initialized correctly**
- ✓ **ServerTick() executes**
- ✓ **View Distance configured**
- ✓ **FOV configured**
- ✓ **Layer masks configured**
- ✓ **Debug visualization enabled**
- ✓ **Visibility volume configured**
- ✓ **Networking integration validated**

Most integration issues can be resolved by checking the items above.

Additional Help

For updated examples, integration guides, and future improvements, refer to the online documentation.

The online documentation is continuously updated and may contain newer information than this PDF version.

15. Frequently Asked Questions (FAQ)

General Questions

What Is The Purpose Of The Anti-Wallhack Visibility System?

The Anti-Wallhack Visibility System is a server-authoritative visibility validation solution designed to reduce information exposure in multiplayer games.

Instead of relying on client-side rendering state, the system determines visibility using physics-based validation, field-of-view checks, and visibility sampling.

This helps prevent wallhack-style exploits and unnecessary network synchronization.

Is This A Networking Framework?

No.

The Anti-Wallhack Visibility System is a visibility validation framework.

It integrates with networking solutions but does not replace them.

Examples:

Photon Fusion Mirror Fish-Networking Netcode for GameObjects Photon PUN

Does The System Replace Interest Management? Not necessarily.

The system can be used alongside interest management solutions or as a visibility layer that helps determine what information should be synchronized.

The exact architecture depends on the requirements of the project.

Networking Questions

Does It Work With Photon Fusion? Yes.

The system was designed with server-authoritative multiplayer architectures in mind and integrates well with Photon Fusion.

The recommended approach is to execute visibility validation inside:

FixedUpdateNetwork()

on the authoritative side.

Does It Work With Mirror?

Yes.

Mirror projects can execute visibility validation on the server and synchronize only the resulting visibility information to clients.

Does It Work With Fish-Networking?

Yes.

Fish-Networking can integrate with the visibility system through the provided adapter architecture.

Does It Work With Netcode For GameObjects?

Yes.

Visibility validation can be executed on the server and used to control synchronization decisions.

Can I Use Another Networking Framework?

Yes.

The visibility system is networking-framework agnostic.

Any multiplayer framework capable of executing server-authoritative logic can integrate with the system.

Visibility Questions

Which Observer Mode Should I Use?

For most multiplayer games:

is recommended.

Benefits:

- Better performance
- Fewer visibility candidates
- More realistic player perception

When Should I Use 360° Mode?

360° mode is useful when directional visibility restrictions are not desired. Examples:

- AI systems
- Top-down games
- RTS games
- Omnidirectional detection

Because more targets pass the observer filtering stage, 360° mode generally has a higher performance cost.

What Is The Difference Between ClosestPoint And BoundsCorners?

ClosestPoint

Benefits:

- Faster
Lower CPU usage

Recommended for:

**Large Player
Counts**

BoundsCorners

Benefits:

- Higher accuracy
- Better edge visibility handling

What Is Velocity Compensation?

Velocity Compensation expands the visibility volume based on movement speed. Benefits:

- Improved corner behavior
- **Better visibility consistency**
- **Fairer multiplayer interactions**

The feature is included in the Free Edition.

Can Velocity Compensation Help With Latency?

Partially.

Increasing velocity compensation can help reduce visibility inconsistencies in higher-latency environments.

However, it should be considered a practical workaround rather than a replacement for dedicated latency-aware visibility systems.

Performance Questions

How Many Players Does The System Support?

The exact number depends on:

- **Visibility settings**
- Update rate
- Sampling density
- Map design
- Hardware

The official benchmark included tests with:

10 Players
25 Player
50 Players
100 Players

using:

32 Updates Per Second 180° Observer Mode Environment Detection Disabled

Real multiplayer games typically distribute players across larger environments and often produce lower workloads than synthetic stress tests.

What Update Rate Should I Use?

Recommended default:

32 TPS

This provides a good balance between responsiveness and performance.

Does Increasing Sampling Density Always Improve Results?

Not necessarily.

Beyond a certain point, increasing sampling density produces diminishing returns while increasing CPU usage.

Start with moderate values and profile before making adjustments.

Integration Questions

Can I Modify The Network Adapter Components?

Yes.

The included adapters are intended as reference implementations.

Most projects will eventually adapt them to match their networking architecture.

Will Asset Updates Overwrite My Adapter Changes?

Potentially.

If the adapter scripts are modified directly, future updates may replace them with newer default versions.

Recommended workflow:

Import Asset ↓ Adapt Components ↓ Move Customized Versions Into Project Code

This helps protect project-specific modifications.

Can I Create My Own Adapter Components?

Absolutely.

The visibility system was designed to support custom integrations.

Many projects create their own adapter layer to match existing architecture and coding standards.

Gameplay Questions

Can I Use The System For PvE Games?

Yes.

The visibility system can be used for:

- **AI perception**
- **Stealth mechanics**
- **NPC awareness**
- **Visibility-based gameplay**

It is not limited to PvP environments.

Can Teammates Always Remain Visible?

Not currently in the Free Edition.

A planned Visibility Filtering system will allow developers to selectively bypass visibility validation for teammates and other friendly entities.

Examples:

Team Markers

Friendly

NPCs Squad

Members Pets

Can I Use The System For Spectators?

Yes.

Custom visibility rules can be implemented through networking integration and future filtering systems.

Support Questions

Where Can I Find Updated Examples?

Framework-specific integration examples are maintained in the online documentation.

The online documentation should always be considered the most up-to-date reference.

The PDF And Online Documentation Are Different. Which One Is Correct?

The online documentation is the primary source of information.

Because the system continues to evolve, the online documentation may contain:

- New examples
- Updated recommendations
- Additional integrations
- New features

before future PDF revisions are published.

Where Should I Start?

Recommended order:

- 1. Quick Start**
- 2. Core Components**
- 3. Network Adapter Components**
- 4. Visibility Pipeline**
- 5. Networking Integration**
- 6. Performance Optimization**

This provides the fastest path from installation to production integration.

If your question is not covered here, refer to the online documentation or future documentation updates for additional guidance.

16. Online Documentation

Overview

This PDF provides an offline reference for the Free Edition of the Anti-Wallhack Visibility System.

However, the online documentation should always be considered the primary and most up-to-date source of information.

Because the system continues to evolve, the online documentation may receive updates before future PDF revisions are published.

Official Documentation

The latest documentation is available at:

<https://gustavobianco277.github.io/Anti-Wallhack-Visibility-System/>

The online version includes:

- Updated setup guides
- Framework integration examples
- Troubleshooting information
- Frequently asked questions
- Performance recommendations
- Benchmark results
- New documentation updates

Why Use The Online Documentation?

Unlike a PDF, the online documentation can be updated immediately whenever:

- New features are released
- Existing features are improved
- Integration examples are updated
- New frameworks are supported
- Best practices evolve

This ensures that developers always have access to the most current information available.

Framework Integration Examples

The online documentation contains framework-specific integration examples that are intentionally omitted from this PDF.

Examples include:

Photon Fusion
Photon PUN 2
Mirror
Fish-Networking
Netcode for
GameObjects

These examples may change over time as networking frameworks receive updates.

For this reason, the online documentation should be consulted whenever implementing or updating multiplayer integrations.

Performance Benchmarks

The latest benchmark results are maintained online.

Benchmark documentation includes:

- Test configurations
- Update rate settings
- Observer configurations
- Performance measurements
- Future benchmark updates

As new versions of the system are released, benchmark information may be expanded and updated.

Future Features

The online documentation may also contain information about upcoming features that are not yet included in the current PDF version.

Examples:

Additional Integrations
New Optimization
Techniques
Future Professional Edition Features

These sections are subject to change as development continues.

Documentation Update Policy

The online documentation is considered the authoritative source for:

Feature Documentation
Integration Examples
Troubleshooting Guides
Optimization
Recommendations Release
Notes

Future PDF revisions will periodically incorporate these updates, but the website should always be treated as the most current reference.

Recommended Workflow

For the best experience:

PDF Documentation
↓
Learn Core Concepts
↓
Online Documentation
↓
Framework Integration
↓
Production Deployment

This approach combines the convenience of an offline reference with the flexibility of continuously updated online resources.

Final Notes

Thank you for using the Anti-Wallhack Visibility System.

The goal of this project is to provide a flexible, scalable, and server-authoritative visibility solution that can be integrated into a wide variety of Unity multiplayer games.

Feedback, suggestions, and future improvements will continue to shape the project as it evolves.

For the latest information, examples, and updates, always refer to the official online documentation.